

A Two-Dimensional Evaluation Framework

External vs Internal Verification of LLM Causal Inference Outputs

RESEARCH QUESTION

When LLMs produce numerical outputs, can we (and they) detect when those outputs are wrong?

CHANNEL A EXTERNAL VERIFICATION

the question it asks
Can a third party
verify correctness?

Verifier: tool / benchmark

Requires: ground truth β^*
(available in benchmarks;
not in deployment)

operationalized by

L2b+

Execute the LLM's R code;
extract coefficient $\hat{\beta}$;
compare to DGP truth β^* ;
pass iff $|\hat{\beta} - \beta^*| / |\beta^*| < 0.5$

empirical result

L2b+ pass rate: 5 - 50%
Ranking $\leftrightarrow$ truth:
Kendall $\tau = +0.73$ [+0.33, +0.87]

✓ WORKS

CHANNEL B INTERNAL VERIFICATION

the question it asks
Can the model itself
signal when wrong?

Verifier: the model itself

Requires: calibrated
self-awareness
(needed in deployment)

operationalized by

Self-Assessment Calibration

Elicit 3-dim confidence
(method / spec / numerical);
compute Expected Calibration
Error (ECE) against L2b+

empirical result

ECE > 0.3 (all models)
 $P(\text{conf} > 0.8 \mid \text{wrong}) >$
 $P(\text{conf} > 0.8 \mid \text{correct})$
→ signal is uninformative

✗ FAILS

THE ASYMMETRY

The channel NEEDED in deployment (internal) is the one that FAILS.
The channel that WORKS (external) requires infrastructure absent in deployment.